Databricks ML Associate

High-Yield Top 100 (Most Frequently Tested Topics)

Practice Exam Deck

How to Use This Deck

100 high-yield questions (Q1–Q100) on the topics that appear most often on the Databricks Machine Learning Associate exam.

Each question slide has four options (A–D); the next slide gives the correct answer and why the others are wrong.

Topic order: Databricks ML platform and AutoML, MLflow, Unity Catalog model registry, Feature Engineering, data processing, tuning, evaluation, deployment.

Exam Domains Covered

Databricks Machine Learning: runtime, AutoML, MLflow, Unity Catalog registry, Feature Engineering client.

Data Processing: EDA, imputation, encoding, Spark ML transformers, pandas API on Spark and UDFs.

Model Development: Hyperopt, CrossValidator, metrics, bias/variance, leakage.

Model Deployment: batch, streaming, real-time serving, monitoring.

Question 1

Which cluster runtime comes preinstalled with MLflow, scikit-learn, XGBoost, and Hyperopt?

A. Photon only
B. Standard Databricks Runtime
C. Databricks Runtime for Machine Learning
D. A SQL warehouse

Answer 1

Correct: C — ML Runtime bundles the common ML libraries and tools.

Wrong:

A: Photon is a faster query engine, not an ML library bundle.

B: does not bundle the ML library set.

D: runs SQL, not notebook ML libraries.

Question 2

A data scientist trains a scikit-learn model on a 2 GB dataset. Which cluster setup is most appropriate?

A. A 16-worker multi-node cluster

B. A SQL warehouse

C. An autoscaling cluster with many workers

D. A Single Node cluster

Answer 2

Correct: D — scikit-learn runs on one machine, so extra workers would sit idle.

Wrong:

A: scikit-learn does not distribute across workers.

B: not meant for Python ML training.

C: workers would still go unused by scikit-learn.

Question 3

Which problem types does Databricks AutoML support?

A. Classification, regression, and clustering
B. Classification, regression, and forecasting
C. Image generation and reinforcement learning
D. Clustering, anomaly detection, and recommendation

Answer 3

Correct: B — These are the three supported AutoML problem types.

Wrong:

A: clustering is not supported.

C: not AutoML problem types.

D: none of these are supported.

Question 4

Why is Databricks AutoML called a glass box" solution?

A. It only works through a no-code UI

B. It automatically deploys the best model to production

C. It generates editable notebooks with the source code for each trial

D. It hides all trial code and returns only the best model

Answer 4

Correct: C — you can inspect, edit, and reuse the generated code.

Wrong:

A: it also has a Python API.

B: AutoML does not auto-deploy.

D: that is a black box.

Question 5

What does AutoML produce that helps you understand data quality before modeling?

A. A serving endpoint
B. A data exploration notebook
C. A Feature Store table
D. A Hyperopt Trials object

Answer 5

Correct: B — it profiles the dataset and flags issues such as nulls and skew.

Wrong:

A: deployment is a separate step.

C: AutoML does not create one for you.

D: this is not an AutoML output.

Question 6

Which call starts an AutoML experiment to predict a numeric target such as price?

A. `databricks.automl.cluster(df)`

B. `mlflow.automl.run(df)`

C. `databricks.automl.regress(df, target_col="price")`

D. `databricks.automl.classify(df, target_col="price")`

Answer 6

Correct: C — regress is for continuous numeric targets.

Wrong:

A: there is no cluster API.

B: not a real API.

D: classify is for categorical targets.

Question 7

After `summary = databricks.automl.classify(...)`, how do you access the best model's details?

A. `summary.champion`

B. `summary.deploy()`

C. `summary.models[0]`

D. `summary.best_trial`

Answer 7

Correct: D — the summary object exposes best_trial with its model path and notebook.

Wrong:

A: not an attribute.

B: AutoML summaries do not deploy.

C: not how the summary is organized.

Question 8

Which Databricks feature syncs notebooks with a Git provider such as GitHub?

A. Workspace trash
B. Git folders (Repos)
C. Unity Catalog volumes
D. DBFS mounts

Answer 8

Correct: B — Git folders provide version control inside the workspace.

Wrong:

A: only holds deleted objects.

C: file storage, not Git.

D: storage access, not version control.

Question 9

What is the correct Unity Catalog naming pattern for a registered model?

A. `workspace.model_name`
B. `schema.model_name`
C. `catalog.schema.model_name`
D. `cluster.catalog.model_name`

Answer 9

Correct: C — UC uses a three-level namespace.

Wrong:

A: no such pattern.

B: missing the catalog level.

D: clusters are not part of the name.

Question 10

A notebook needs a library that is not on the cluster. Which command installs it only for the current notebook session?

A. Editing the cluster policy

B. `spark.conf.set()`

C. `dbutils.fs.cp()`

D. `%pip install <package>`

Answer 10

Correct: D — notebook-scoped installs affect only that notebook's environment.

Wrong:

A: this does not install packages.

B: sets Spark config, not libraries.

C: copies files.

Question 11

How should an API token be used in a notebook without exposing it in code?

A. Store it in a markdown cell

B. Store it in a secret scope and read it with `dbutils.secrets.get()`

C. Hard-code it in a variable

D. Put it in a notebook widget default

Answer 11

Correct: B — secrets are redacted and access-controlled.

Wrong:

A: plain text and visible.

C: exposes the credential.

D: widgets are visible.

Question 12

Where should you store unstructured files (such as images) under Unity Catalog governance?

A. A feature table
B. A Delta table column of file names only
C. The cluster's local disk
D. A Unity Catalog volume

Answer 12

Correct: D — volumes govern non-tabular files.

Wrong:

A: for features, not files.

B: does not store or govern the files.

C: ephemeral and ungoverned.

Question 13

A team wants a notebook to retrain a model every night at lower cost. What should they use?

A. A Git folder sync

B. A SQL alert

C. A Databricks Workflows job with a schedule on a job cluster

D. Run manually on an all-purpose cluster

Answer 13

Correct: C — job clusters start for the run and terminate afterward.

Wrong:

A: does not run notebooks.

B: alerts do not retrain models.

D: not automated and costs more.

Question 14

How can a team reproduce the exact training data used for an earlier model version in Delta Lake?

A. Run `OPTIMIZE`

B. Re-run `df.cache()`

C. Use Delta time travel with `versionAsOf` or `timestampAsOf`

D. Run `VACUUM` with 0 hours

Answer 14

Correct: C — time travel reads earlier table versions.

Wrong:

A: compacts files.

B: caching is not versioning.

D: deletes old files and breaks time travel.

Question 15

What is an MLflow experiment?

A. A single training execution
B. A Delta table of features
C. A deployed REST endpoint
D. A named collection of runs for a particular task

Answer 15

Correct: D — runs are grouped under an experiment.

Wrong:

A: that is a run.

B: that is a feature table.

C: that is Model Serving.

Question 16

Which code correctly creates a tracked run in a chosen experiment?

A. `mlflow.set_experiment(path)` then `with mlflow.start_run():`

B. `mlflow.new_run(path)`

C. `mlflow.log_run()` alone

D. `mlflow.create_experiment_run()`

Answer 16

Correct: A — set the experiment, then open a run context.

Wrong:

B: not an API.

C: does not start a run.

D: not an API.

Question 17

Which call logs a value that can be recorded repeatedly with a step, such as training loss per epoch?

A. `mlflow.log_metric("loss", value, step=epoch)`

B. `mlflow.set_tag("loss", value)`

C. `mlflow.log_param("loss", value)`

D. `mlflow.log_artifact("loss")`

Answer 17

Correct: A — metrics are numeric and support steps.

Wrong:

B: tags are string metadata.

C: params are fixed once set.

D: logs files.

Question 18

Which call saves a plot image file to a run?

A. `mlflow.set_tag("plot.png")`
B. `mlflow.log_artifact("plot.png")`
C. `mlflow.log_metric("plot.png")`
D. `mlflow.log_param("plot.png")`

Answer 18

Correct: B — artifacts are files and directories attached to a run.

Wrong:

A: tags are text labels.

C: metrics are numbers.

D: params are small key-value pairs.

Question 19

What is true about Databricks Autologging on ML Runtime clusters?

A. It works only for Spark ML
B. It is enabled by default for supported libraries
C. It only works after `mlflow.register_model()`
D. It must be set up with an init script

Answer 19

Correct: B — supported flavors log params, metrics, and models automatically.

Wrong:

A: many libraries are supported.

C: unrelated.

D: no init script is needed.

Question 20

What does `mlflow.search_runs()` return by default?

A. A registered model version
B. A single best model
C. A Spark Streaming query
D. A pandas DataFrame of runs

Answer 20

Correct: D — it returns a DataFrame you can sort and filter.

Wrong:

A: it does not touch the registry.

B: it returns run records.

C: not what it returns.

Question 21

Which call returns the run with the lowest RMSE in an experiment?

A. `mlflow.search_runs(experiment_ids=[id], order_by=["metrics.rmse ASC"], max_results=1)`

B. `mlflow.get_min_run()`

C. `mlflow.best_run("rmse")`

D. `mlflow.search_runs(order_by=["metrics.rmse DESC"], max_results=1)`

Answer 21

Correct: A — sort by the metric ascending and keep one row.

Wrong:

B: not an API.

C: not an API.

D: this returns the highest RMSE.

Question 22

What does an MLflow model signature describe?

A. The cluster used for training
B. The input and output schema of the model
C. The model's license
D. The hyperparameter search space

Answer 22

Correct: B — it records expected column names and types.

Wrong:

A: not part of the signature.

C: not part of the signature.

D: not part of the signature.

Question 23

Which function infers a model signature from data?

A. `mlflow.models.deploy(X)`

B. `mlflow.sign(X)`

C. `mlflow.schema.get(X)`

D. `mlflow.models.infer_signature(X, predictions)`

Answer 23

Correct: D — it derives schema from sample inputs and outputs.

Wrong:

A: deployment is unrelated.

B: not an API.

C: not an API.

Question 24

Which MLflow flavor lets you load any logged model and call a generic `predict()`?

A. `mlflow.tracking`
B. `python_function` (pyfunc)
C. `mlflow.projects`
D. `mlflow.spark` only

Answer 24

Correct: B — pyfunc is the generic, framework-agnostic interface.

Wrong:

A: is the tracking API.

C: packages code, not models.

D: limited to Spark ML models.

Question 25

How do you apply a logged MLflow model to a large Spark DataFrame?

A. `df.toPandas()` then predict on the driver

B. `mlflow.pyfunc.spark_udf(spark, model_uri)`

C. `model.collect()`

D. `mlflow.score(df)`

Answer 25

Correct: B — it wraps the model as a Spark UDF for distributed scoring.

Wrong:

A: does not scale.

C: models have no such method.

D: not an API.

Question 26

A team tunes hyperparameters and wants one parent run with a child run per trial. What should they use?

A. `mlflow.end_run()` repeatedly

B. Separate experiments per trial

C. `mlflow.log_param` for all trials in one run

D. `mlflow.start_run(nested=True)` inside a parent run

Answer 26

Correct: D — nested runs create child runs under a parent.

Wrong:

A: ends runs, not nesting them.

B: loses the grouping.

C: does not create child runs.

Question 27

How can you visually compare many runs in the MLflow UI?

A. Select runs and click Compare to see parallel coordinates and scatter plots

B. Open each run's artifact folder

C. Check the cluster event log

D. Use the Spark UI

Answer 27

Correct: A — the compare view helps relate parameters to metrics.

Wrong:

B: slow and not comparative.

C: unrelated to run comparison.

D: shows jobs, not runs.

Question 28

How do you register a model at the time you log it?

A. Pass `registered_model_name=` to `mlflow.<flavor>.log_model()`

B. Pass `register=True` to `mlflow.start_run()`

C. Call `mlflow.sklearn.save_model()`

D. Use `mlflow.log_param("register", True)`

Answer 28

Correct: A — log_model can create the registered model version directly.

Wrong:

B: not a parameter.

C: saves locally only.

D: only logs a parameter.

Question 29

Which call registers a model from an existing run?

A. `mlflow.sklearn.save_model(run_id)`

B. `mlflow.create_model(run_id)`

C. `mlflow.log_metric("register", 1)`

D. `mlflow.register_model("runs:/<run_id>/model", "catalog.schema.name")`

Answer 29

Correct: D — it creates a model version from the run's model artifact.

Wrong:

A: saves locally.

B: not an API.

C: logs a metric only.

Question 30

Which URI format loads a model directly from a run?

A. `dbfs:/<run_id>`

B. `runs:/<run_id>/model`

C. `experiments:/<id>/model`

D. `run://model/<run_id>`

Answer 30

Correct: B — this is the run-based model URI.

Wrong:

A: points to storage, not the model URI scheme.

C: not a model URI scheme.

D: not valid.

Question 31

Which call points MLflow at the Unity Catalog model registry?

A. `mlflow.set_registry_uri("uc://")`

B. `mlflow.set_registry_uri("databricks")`

C. `mlflow.set_tracking_uri("unity")`

D. `mlflow.set_registry_uri("databricks-uc")`

Answer 31

Correct: D — this selects UC as the registry.

Wrong:

A: not valid.

B: selects the legacy workspace registry.

C: tracking URI is not the registry.

Question 32

What replaces the legacy Staging/Production stages in the Unity Catalog model registry?

A. Aliases and tags
B. Experiment folders
C. Git branches
D. Cluster tags

Answer 32

Correct: A — UC uses aliases such as Champion to mark versions.

Wrong:

B: do not label versions.

C: not part of the registry.

D: label compute, not models.

Question 33

Which URI loads the model version that currently has the alias Champion?

A. `runs:/Champion`
B. `models:/main.ml.churn/Champion`
C. `models:/main.ml.churn@Champion`
D. `models:/Champion/main.ml.churn`

Answer 33

Correct: C — the `@` syntax resolves an alias.

Wrong:

A: not a run ID.

B: legacy stage syntax.

D: wrong order.

Question 34

Which call assigns the alias Champion to version 3 of a UC model?

A. `client.transition_model_version_stage(name, 3, "Champion")`

B. `client.register_alias(3)`

C. `MlflowClient().set_registered_model_alias(name, "Champion", 3)`

D. `mlflow.set_alias(name, 3)`

Answer 34

Correct: C — this sets an alias on a specific version.

Wrong:

A: stages are legacy.

B: not an API.

D: not an API.

Question 35

Production loads `models:/main.ml.churn@Champion`. A new version is validated. How do you promote it with no code change?

A. Edit every notebook with the new version number
B. Delete the registered model
C. Reassign the Champion alias to the new version
D. Rename the experiment

Answer 35

Correct: C — clients resolve the alias at load time.

Wrong:

A: unnecessary work.

B: breaks consumers.

D: has no effect on aliases.

Question 36

Which privilege lets a group load and call a registered model in Unity Catalog (in addition to USE CATALOG and USE SCHEMA)?

A. `SELECT` on the model

B. `CREATE TABLE` on the schema

C. `EXECUTE` on the model

D. `MODIFY` on the cluster

Answer 36

Correct: C — EXECUTE allows using the model.

Wrong:

A: applies to tables.

B: for tables.

D: unrelated.

Question 37

Which client is used for feature tables governed by Unity Catalog?

A. `FeatureEngineeringClient`
B. `MlflowClient`
C. `FeatureStoreClient` (workspace only)
D. `SparkSession`

Answer 37

Correct: A — from `databricks.feature_engineering`.

Wrong:

B: manages tracking and registry.

C: the older workspace feature store client.

D: is Spark, not a feature client.

Question 38

What does a feature table require?

A. An MLflow run
B. A serving endpoint
C. A GPU cluster
D. A primary key

Answer 38

Correct: D — rows are identified by the primary key for lookups and merges.

Wrong:

A: not required.

B: not required.

C: not required.

Question 39

Which call creates a feature table from a DataFrame?

A. `fe.create_table(name=..., primary_keys=["customer_id"], df=df)`

B. `fe.new_features(df)`

C. `fe.log_model(df)`

D. `fe.score_batch(df)`

Answer 39

Correct: A — it defines the key and initial data.

Wrong:

B: not an API.

C: logs a model.

D: scores data.

Question 40

Why add a timestamp key to a feature table?

A. To enable point-in-time lookups and avoid leakage
B. To encrypt the table
C. To replace the primary key
D. To make training faster

Answer 40

Correct: A — training rows join to the latest feature values as of their timestamp.

Wrong:

B: not its purpose.

C: it is used together with it.

D: not its purpose.

Question 41

What is `FeatureLookup` used for?

A. Specifying which features to join to a label DataFrame by key
B. Starting an AutoML run
C. Registering a model
D. Creating a cluster

Answer 41

Correct: A — it names the table, lookup key, and features.

Wrong:

B: not its role.

C: not its role.

D: not its role.

Question 42

Which call joins feature lookups to a label DataFrame to build training data?

A. `mlflow.search_runs(df)`

B. `fe.score_batch(df)`

C. `fe.create_training_set(df, feature_lookups, label)` and then `.load_df()`

D. `fe.write_table(df)`

Answer 42

Correct: C — it assembles features and labels for training.

Wrong:

A: searches runs.

B: performs inference.

D: writes feature data.

Question 43

Why log a model with `fe.log_model()` instead of a plain MLflow flavor?

A. It disables autologging

B. It skips the registry

C. It stores feature lookup metadata so features are retrieved automatically at inference

D. It makes the model smaller

Answer 43

Correct: C — this prevents training-serving skew.

Wrong:

A: not its purpose.

B: not its purpose.

D: not its purpose.

Question 44

A model was logged with `fe.log_model()`. At scoring time the input has only `customer_id`. Which call should be used?

A. `mlflow.search_runs(df)`

B. `fe.create_table(df)`

C. `fe.score_batch(model_uri=..., df=df)`

D. `model.predict(df)` with the raw key

Answer 44

Correct: C — it looks up the needed features automatically.

Wrong:

A: not scoring.

B: creates a table.

D: missing features would cause an error.

Question 45

Which write mode updates existing feature rows by primary key and inserts new ones?

A. `mode="merge"`

B. `mode="append"`

C. `mode="delete"`

D. `mode="overwrite"`

Answer 45

Correct: A — merge performs an upsert on the key.

Wrong:

B: not the supported behavior here.

C: not a write mode.

D: replaces the whole table.

Question 46

What is a core benefit of using a feature store?

A. Automatic hyperparameter tuning

B. Free GPU time

C. Removing the need for labels

D. Reusing consistent features across training and inference with lineage

Answer 46

Correct: D — this reduces duplicate work and skew.

Wrong:

A: that is a tuning tool.

B: not a benefit.

C: labels are still required.

Question 47

A model needs low-latency feature lookups at real-time inference. What is typically required?

A. Publish the feature table to an online store (online table)

B. Read the offline Delta table on every request

C. Disable the primary key

D. Store features in a notebook variable

Answer 47

Correct: A — online stores serve low-latency lookups.

Wrong:

B: too slow.

C: keys are needed.

D: not durable.

Question 48

Which Databricks utility produces summary statistics and distributions for a DataFrame?

A. `dbutils.notebook.run(df)`

B. `dbutils.data.summarize(df)`

C. `dbutils.widgets.get(df)`

D. `dbutils.fs.ls(df)`

Answer 48

Correct: **B** — it renders a data profile.

Wrong:

A: runs notebooks.

C: reads widgets.

D: lists files.

Question 49

Which DataFrame method returns count, mean, stddev, min, quartiles, and max?

A. `df.summary()`

B. `df.describe()`

C. `df.explain()`

D. `df.printSchema()`

Answer 49

Correct: A — summary includes percentiles, unlike describe.

Wrong:

B: omits the quartiles by default.

C: shows the query plan.

D: shows column types.

Question 50

Which statistic is most robust for imputing a skewed numeric feature?

A. Median

B. Maximum

C. Mean

D. Sum

Answer 50

Correct: A — the median is not pulled by extreme values.

Wrong:

B: distorts the distribution.

C: is pulled by outliers.

D: is not a valid imputation value.

Question 51

Which strategies does `pyspark.ml.feature.Imputer` support?

A. forward-fill and back-fill

B. regression and KNN

C. mean, max, and min

D. mean, median, and mode

Answer 51

Correct: D — these are the available strategies.

Wrong:

A: not supported.

B: not supported.

C: max and min are not strategies.

Question 52

A team wants to preserve information about which values were missing. What should they add?

A. Drop the column

B. Impute and discard the original

C. Cast all values to string

D. A boolean indicator column for missingness before imputing

Answer 52

Correct: D — the model can learn from the missingness pattern.

Wrong:

A: loses information.

B: loses missingness information.

C: does not help.

Question 53

A data scientist calls `transform()` directly on a new `Imputer` and gets an error. Why?

A. `transform()` needs a GPU

B. `Imputer` is an estimator and must be fit first

C. `Imputer` requires a feature table

D. `Imputer` only works on strings

Answer 53

Correct: B — `fit()` returns an `ImputerModel`, which has `transform()`.

Wrong:

A: not true.

C: not true.

D: it works on numeric columns.

Question 54

Which is the correct Spark ML order to one-hot encode a string column?

A. `VectorAssembler` then `StringIndexer`
B. `Imputer` then `OneHotEncoder` only
C. `OneHotEncoder` then `StringIndexer`
D. `StringIndexer` then `OneHotEncoder`

Answer 54

Correct: D — strings must be indexed to numbers before encoding.

Wrong:

A: assembler builds the final vector.

B: strings are not indexed.

C: encoder needs numeric indices.

Question 55

What does `StringIndexer(handleInvalid="keep")` do for labels unseen during training?

A. Raises an error

B. Deletes the column

C. Maps them to the mean

D. Assigns them an extra index instead of failing

Answer 55

Correct: D — this avoids errors on new categories at inference.

Wrong:

A: that is the default `error`.

B: not the behavior.

C: not the behavior.

Question 56

What does `VectorAssembler` do?

A. Scales features to unit variance

B. Trains a model

C. Splits a vector into columns

D. Combines selected columns into a single vector column

Answer 56

Correct: D — Spark ML models expect one features vector.

Wrong:

A: that is `StandardScaler`.

B: it is a transformer.

C: the opposite.

Question 57

What does `Pipeline.fit()` return?

A. A pandas DataFrame

B. A `Pipeline`

C. A `CrossValidator`

D. A `PipelineModel`

Answer 57

Correct: D — fitting estimators produces fitted transformers chained as one model.

Wrong:

A: not what it returns.

B: that is the unfitted object.

C: not what it returns.

Question 58

Which model type generally needs feature scaling the least?

A. Tree-based models such as random forests
B. Neural networks
C. Linear regression with regularization
D. k-nearest neighbors

Answer 58

Correct: A — splits depend on order, not magnitude.

Wrong:

B: typically benefit from scaling.

C: regularization is scale sensitive.

D: uses distances and benefits from scaling.

Question 59

A model is trained on log(price). What must be done before computing RMSE against original prices?

A. Back-transform predictions with exp

B. Nothing, compare directly

C. Drop the highest predictions

D. Take the log of the original prices twice

Answer 59

Correct: A — metrics must compare values on the same scale.

Wrong:

B: the scales differ.

C: not a fix.

D: wrong transformation.

Question 60

What is lazy evaluation in Spark?

A. All operations run immediately
B. Only SQL is optimized
C. Transformations are not run until an action is called
D. Data is deleted after each step

Answer 60

Correct: C — actions such as `count()` trigger execution.

Wrong:

A: that is eager.

B: not true.

D: not true.

Question 61

Which call converts a Spark DataFrame to a pandas API on Spark DataFrame, keeping it distributed?

A. `df.pandas_api()`

B. `df.collect()`

C. `df.toPandas()`

D. `df.cache()`

Answer 61

Correct: A — it returns a distributed pandas-style DataFrame.

Wrong:

B: returns a list on the driver.

C: collects everything to the driver.

D: only caches.

Question 62

Why is `toPandas()` risky on a very large DataFrame?

A. It converts to RDD only

B. It collects all data to the driver and may run out of memory

C. It deletes the Spark table

D. It always runs faster

Answer 62

Correct: B — the driver holds the entire result.

Wrong:

A: not true.

C: not true.

D: not true.

Question 63

Why are pandas UDFs generally faster than row-at-a-time Python UDFs?

A. They skip serialization
B. They use GPUs by default
C. They process batches using Apache Arrow
D. They run only on the driver

Answer 63

Correct: C — vectorized execution avoids per-row overhead.

Wrong:

A: Arrow still transfers data.

B: not by default.

D: they run on workers.

Question 64

A team wants to train one model per store in parallel across a cluster. What should they use?

A. `df.coalesce(1)`

B. `df.groupBy("store").applyInPandas(train_fn, schema)`

C. `df.toPandas()` and a for-loop

D. `df.limit(1)`

Answer 64

Correct: B — each group is processed as a pandas DataFrame on workers.

Wrong:

A: forces one partition.

C: runs on the driver.

D: keeps one row.

Question 65

Which call performs a reproducible random train/test split of a Spark DataFrame?

A. `train_test_split(df)` from scikit-learn on a Spark DataFrame

B. `df.split(0.8)`

C. `df.limit(0.8)`

D. `df.randomSplit([0.8, 0.2], seed=42)`

Answer 65

Correct: D — the seed makes the split repeatable.

Wrong:

A: expects pandas or arrays.

B: not an API.

C: limit takes a row count.

Question 66

Which step is an example of data leakage?

A. Using a fixed random seed
B. Computing normalization statistics on the full dataset before splitting
C. Splitting before preprocessing
D. Fitting the scaler on training data only

Answer 66

Correct: B — test information leaks into training.

Wrong:

A: this aids reproducibility.

C: this is correct practice.

D: this is correct practice.

Question 67

A dataset has 1 percent positive labels. Which split technique keeps class proportions?

A. Sorting by label

B. Stratified splitting

C. Using only the first rows

D. Splitting by column order

Answer 67

Correct: B — stratification preserves the label ratio.

Wrong:

A: creates biased splits.

C: is not representative.

D: ignores class balance.

Question 68

How should a time series dataset be split for evaluation?

A. Train on earlier data and test on later data without shuffling

B. Random shuffle split

C. Use the same rows for train and test

D. Train on the latest and test on the earliest

Answer 68

Correct: A — this mimics real forecasting and avoids future leakage.

Wrong:

B: leaks future information.

C: no unseen test data.

D: reverses time.

Question 69

Which technique helps with a heavily imbalanced classification dataset?

A. Using accuracy as the only metric

B. Class weights or resampling the training data

C. Duplicating the test set

D. Removing the minority class

Answer 69

Correct: B — these reduce bias toward the majority class.

Wrong:

A: accuracy is misleading.

C: contaminates evaluation.

D: destroys the signal.

Question 70

Hyperopt `fmin` minimizes the objective. How do you maximize accuracy?

A. Return accuracy directly

B. Return the negative accuracy as the loss

C. Use `algo=max.suggest`

D. Set `maximize=True`

Answer 70

Correct: B — minimizing -accuracy maximizes accuracy.

Wrong:

A: this would minimize accuracy.

C: not an algorithm.

D: no such argument.

Question 71

When is `SparkTrials` an appropriate choice with Hyperopt?

A. When using no objective function
B. When tuning only on the driver
C. When tuning single-machine libraries such as scikit-learn
D. When tuning distributed Spark ML models

Answer 71

Correct: C — each trial runs on a worker in parallel.

Wrong:

A: an objective is required.

B: SparkTrials uses workers.

D: those already use the cluster.

Question 72

What does the `parallelism` setting in `SparkTrials` control?

A. The number of hyperparameters
B. The model's tree depth
C. The maximum number of trials evaluated concurrently
D. The number of folds

Answer 72

Correct: C — more parallelism is faster but makes TPE less adaptive.

Wrong:

A: not controlled here.

B: not controlled here.

D: that is for cross-validation.

Question 73

Which Hyperopt space is best for an integer-like parameter such as `max_depth`?

A. `hp.uniform("max_depth", 2, 20)`

B. `hp.quniform("max_depth", 2, 20, 1)` with an int cast

C. `hp.pyll("max_depth")`

D. `hp.normal("max_depth", 0, 1)`

Answer 73

Correct: B — quniform returns rounded floats, so cast to int.

Wrong:

A: returns arbitrary floats.

C: not an API.

D: wrong distribution.

Question 74

`fmin` is run with `hp.choice` for a parameter. What does the returned best dict contain for it?

A. A Spark DataFrame

B. The loss value

C. The actual chosen value

D. The index of the chosen option

Answer 74

Correct: D — use `space_eval` to recover the actual value.

Wrong:

A: not what it returns.

B: not for the parameter.

C: not for hp.choice.

Question 75

How does `tpe.suggest` differ from `rand.suggest`?

A. TPE only works for Spark ML
B. TPE samples uniformly at random
C. TPE uses past trial results to choose promising values (Bayesian-style)
D. TPE does not need an objective

Answer 75

Correct: C — random search ignores previous results.

Wrong:

A: it works for any objective.

B: that is rand.suggest.

D: it needs one.

Question 76

A 3-fold CrossValidator uses a grid of 4 x 3 = 12 parameter combinations. How many models are trained during the search (excluding the final refit)?

A. 36

B. 12

C. 15

D. 3

Answer 76

Correct: A — 12 combinations times 3 folds.

Wrong:

B: ignores the folds.

C: adds instead of multiplying.

D: counts folds only.

Question 77

How does `TrainValidationSplit` differ from `CrossValidator`?

A. It uses k folds

B. It only works for clustering

C. It evaluates each parameter combination once on a single split

D. It tunes without an evaluator

Answer 77

Correct: C — it is cheaper but less reliable.

Wrong:

A: that is `CrossValidator`.

B: not true.

D: an evaluator is needed.

Question 78

Which three things must be supplied when constructing a Spark ML `CrossValidator`?

A. feature table, lookup, and label
B. model, registry, and endpoint
C. cluster, runtime, and job
D. estimator, estimatorParamMaps, and evaluator

Answer 78

Correct: D — the estimator, grid, and metric define the search.

Wrong:

A: unrelated.

B: unrelated.

C: unrelated.

Question 79

A Spark ML decision tree fails because a categorical feature has more categories than allowed. What should be changed?

A. Increase `numTrees` only

B. Increase `maxBins`

C. Disable the pipeline

D. Decrease `maxDepth`

Answer 79

Correct: B — maxBins must be at least the number of categories.

Wrong:

A: does not address bins.

C: not a fix.

D: does not address bins.

Question 80

A fraud dataset has 1 percent positives. Why is accuracy a poor metric?

A. Accuracy ignores true negatives

B. Predicting all negatives gives 99 percent accuracy

C. Accuracy cannot be computed

D. Accuracy is only for regression

Answer 80

Correct: B — accuracy hides failure on the rare class.

Wrong:

A: it includes them.

C: it can be computed.

D: it is for classification.

Question 81

A screening model must catch as many true positives as possible. Which metric should be maximized?

A. Recall

B. Precision

C. RMSE

D. Silhouette score

Answer 81

Correct: A — recall measures the share of positives found.

Wrong:

B: measures correctness of positive predictions.

C: a regression metric.

D: a clustering metric.

Question 82

What does ROC AUC represent?

A. The fraction of correct predictions

B. The number of features used

C. The average squared error

D. The probability that a random positive is ranked above a random negative

Answer 82

Correct: D — it measures ranking quality across thresholds.

Wrong:

A: that is accuracy.

B: not related.

C: that is MSE.

Question 83

RMSE is much larger than MAE for a regression model. What does this suggest?

A. The model is perfect

B. A few large errors are present

C. There are no outliers

D. The features are scaled

Answer 83

Correct: B — RMSE squares errors and penalizes large ones more.

Wrong:

A: errors are present.

C: outliers are likely.

D: unrelated.

Question 84

A regression model has R2 close to 0 on the test set. What does this mean?

A. It has zero error

B. It explains all variance

C. It performs about as well as predicting the mean

D. It is perfectly calibrated

Answer 84

Correct: C — R2 compares the model to a mean baseline.

Wrong:

A: not implied.

B: that is R2 near 1.

D: not implied.

Question 85

Training accuracy is 99 percent and validation accuracy is 72 percent. What is the likely issue and a fix?

A. Label leakage in serving

B. Underfitting; add more layers

C. Overfitting; add regularization, simplify the model, or get more data

D. Data drift; redeploy

Answer 85

Correct: C — a large train/validation gap indicates overfitting.

Wrong:

A: not indicated.

B: underfitting shows poor training performance.

D: not indicated.

Question 86

Which statement correctly contrasts bagging and boosting?

A. Boosting trains independent models in parallel

B. Bagging mainly reduces variance; boosting builds models sequentially to reduce bias

C. Bagging builds models sequentially

D. Both only reduce bias

Answer 86

Correct: B — random forests bag; gradient boosting is sequential.

Wrong:

A: that is bagging.

C: that is boosting.

D: bagging mainly reduces variance.

Question 87

Which regularization can shrink some coefficients exactly to zero?

A. Dropout on linear models
B. No regularization
C. L1 (Lasso)
D. L2 (Ridge)

Answer 87

Correct: C — L1 produces sparse solutions and acts as feature selection.

Wrong:

A: not the standard answer.

B: no shrinkage.

D: shrinks but rarely zeroes coefficients.

Question 88

What is the default metric of Spark ML `RegressionEvaluator`?

A. areaUnderROC

B. r2

C. f1

D. rmse

Answer 88

Correct: D — rmse is the default metric name.

Wrong:

A: belongs to the binary classification evaluator.

B: available but not the default.

C: belongs to the multiclass evaluator.

Question 89

Which Spark ML evaluator computes areaUnderROC?

A. `BinaryClassificationEvaluator`
B. `RankingEvaluator`
C. `ClusteringEvaluator`
D. `RegressionEvaluator`

Answer 89

Correct: A — its default metric is areaUnderROC.

Wrong:

B: not the answer here.

C: clustering metrics.

D: regression metrics.

Question 90

What is the difference between a Spark ML estimator and a transformer?

A. Estimators transform data directly

B. They are the same class

C. An estimator has `fit()` and returns a model; a transformer has `transform()`

D. Transformers have `fit()` only

Answer 90

Correct: C — fit produces a fitted transformer.

Wrong:

A: they must be fit first.

B: they are distinct.

D: they have `transform()`.

Question 91

How can you inspect which features a fitted random forest relies on most?

A. Look at the learning rate
B. Look at the cluster size
C. Look at the run ID
D. Look at feature importances

Answer 91

Correct: D — tree ensembles expose importances (`featureImportances` in Spark ML).

Wrong:

- A: not importances.
- B: not model-related.
- C: not importances.

Question 92

Which technique explains an individual prediction by feature contribution?

A. Elbow plot

B. SHAP values

C. ROC curve

D. Confusion matrix

Answer 92

Correct: B — SHAP attributes a prediction to features.

Wrong:

A: used for choosing k.

C: summarizes threshold performance.

D: summarizes classification outcomes.

Question 93

A model must score 100 million Delta records overnight. Which approach is best?

A. `toPandas()` then loop on the driver

B. Manual scoring in a spreadsheet

C. Batch inference with Spark, for example `spark_udf` or `fe.score_batch()`

D. A real-time REST endpoint per row

Answer 93

Correct: C — batch scoring is efficient for large offline jobs.

Wrong:

A: does not scale.

B: not feasible.

D: too slow and costly.

Question 94

A team needs low-latency REST predictions. What should they create?

A. A Git folder
B. A Model Serving endpoint
C. A SQL dashboard
D. A scheduled batch job

Answer 94

Correct: B — serving endpoints expose models as REST APIs.

Wrong:

A: not deployment.

C: not a model API.

D: not low latency.

Question 95

A team needs continuous scoring as new data arrives. What is a suitable approach?

A. Spark Structured Streaming with a model UDF

B. Exporting to CSV weekly

C. A one-time `OPTIMIZE`

D. Manually re-running a notebook hourly

Answer 95

Correct: A — streaming applies the model to incoming micro-batches.

Wrong:

B: not streaming.

C: unrelated.

D: not continuous.

Question 96

Which request key is valid when sending data to a Databricks serving endpoint as JSON?

A. `model_input_df`

B. `records_pandas`

C. `dataframe_records`

D. `spark_rows`

Answer 96

Correct: C — other valid formats include `dataframe_split` and `inputs`.

Wrong:

A: not valid.

B: not valid.

D: not valid.

Question 97

A team wants to split traffic between two model versions on one endpoint. What should they configure?

A. Traffic percentages across served entities

B. Two separate experiments

C. Different Git branches

D. Two feature tables

Answer 97

Correct: A — an endpoint can serve multiple versions with a traffic split.

Wrong:

B: unrelated.

C: unrelated.

D: unrelated.

Question 98

What do inference tables capture for a serving endpoint?

A. Notebook revisions
B. Request and response payloads logged to a Delta table
C. Training data versions
D. Cluster logs only

Answer 98

Correct: B — they support monitoring and debugging.

Wrong:

A: not their purpose.

C: not their purpose.

D: not their purpose.

Question 99

What is the difference between data drift and concept drift?

A. Data drift is a change in labels only

B. Concept drift only happens during training

C. They are identical terms

D. Data drift is a change in input distributions; concept drift is a change in the feature-target relationship

Answer 99

Correct: D — both can degrade performance in production.

Wrong:

A: it concerns inputs.

B: it happens after deployment.

C: they differ.

Question 100

Which Databricks capability monitors tables and inference data for drift and quality?

A. Git folders

B. `dbutils.fs`

C. Cluster autoscaling

D. Lakehouse Monitoring

Answer 100

Correct: D — it tracks statistics and drift over time.

Wrong:

A: version control.

B: file utilities.

C: compute scaling.